

AZURE NETWORKING FOR THE DATA PROFESSIONAL

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HI, I'M DREW

It's nice to meet you



- Solutions Architect, Data Engineer, Dog Person -



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2018-2020

Professional Skills

Here's what I'm good at and enjoy



About Me

Some things about me that you should know

I'm a Senior Solutions Architect. Before that, I was a SQL Server DBA since the days of SQL Server 2000 and it's still my first love. I also enjoy DevOps, data engineering, release management, and, of course, Microsoft Azure.

I also like to do fun projects like the High Altitude SQL Server Project (HASSP) or playing with my dog. I also haven't seen any of the "Avengers" movies so I'd appreciate it if you didn't spoil them for me.

I also have a small, tailor-made consulting company, Dublin Data Solutions, that can also assist in your cloud migration or data modernization plans. Contact me if you'd like to know more!

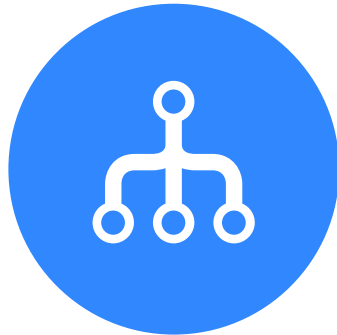
WHY YOU NEED THIS

- Raise your hand if you've ever blamed the network
- When you're working Azure (or any cloud provider, really) it's important that *everyone* in your organization understands how your services are being connected to
- This means that you should know:
 - Understand your cloud network topology and how the services you're responsible for live in it
 - What a private link is, and why you (probably) need it
 - If your services are available publicly or privately (and how to tell the difference)
 - Where to look if someone can't connect

WHAT TO EXPECT FROM THIS COURSE

Networking 101

An introduction to software defined networking in Azure



Azure Services and Access

In which we discuss public vs. private access to your services, and how to know the differences

Azure Private Link

Complete control of your service access to only the networks you want, even from on-premises



DNS Resolution in Azure

Obligatory “It’s always DNS” joke here



On-Premises Connectivity

How to (securely) connect to Azure to access your already secure services



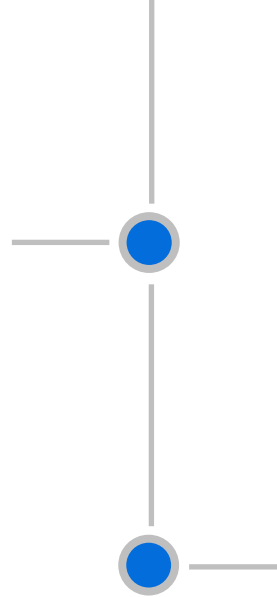
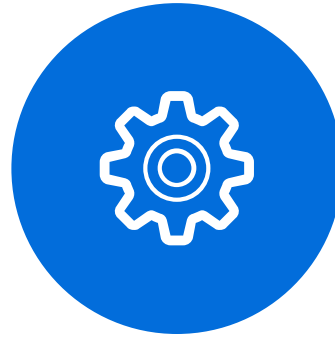
Common Topologies

Look at some common network deployment strategies, and how you can be a good citizen of them



Managed Virtual Networks

Some services connect securely different than others. We'll look at one of them by way of an example



Closing

Where to go or learn next, and what's coming soon

IT'S MILLER TIME™

FOLLOWING ALONG ON YOUR OWN

- This course will be working in Azure a lot, obviously
- You may or may not have access to do the same things the demos do, like creating resources
- This isn't strictly required; you can watch the demos and learn by watch me spend money instead
- However, if you want the full effect, you can:
 - Ask your cloud administrators for a learning subscription that you have full access to
 - Create your own Azure account (it's free*) and comes with \$200.00 in credit** and offers some cool services for free for the entire term (with limits, of course)
 - <https://azure.microsoft.com/en-us/free/>
 - * You still need a credit card to sign up
 - ** You need to use (or lose) the credit in your first 30 days
- Delete resources when you're done!



NETWORKING 101

Some common terms and technologies you'll need to be familiar with

WE'RE ALL NETWORK ADMINS NOW

- Before we dive into some of the topics of this course, we need to level-set on some terms and technologies.
- Just like regular networks, the goal is to have IP Addresses to assign to services, virtual machines or other things in Azure
- You'll be seeing these terms a lot throughout the course
 - **Virtual Network (VNet):** A software-defined network that describes a network. Compared to a regular, traditional network, there's no wires to manage or ports to configure. The VNet definition and configuration takes care of this for us. A VNet can span practically any distance to support devices or services anywhere. A virtual network is defined first by the address space (size), which is specified in **Classless Inter-Domain Routing (CIDR) notation** (more on that in a moment). This address space specifies the range of IPs that can exist within it.
 - **Subnet:** Since VNets can span very large address space, it's common practice to segment, or divide them, into smaller portions for ease of management, traffic isolation, or delegation. Subnets are defined an existing VNet and are also specified by CIDR notation. Each VNet must have at least one subnet defined and can have as many as they'd like if there are available IP ranges to assign.

CLASSLESS INTER-DOMAIN ROUTING (CIDR)

- Classless Inter-domain routing is a very common notation for defining the address space (available IPs) of a network and subnets within them.
- You don't need to memorize this, but you *should* know how to read it
- Format: Looks like an IP address with a slash and number at the end
- The IP number represents the first IP address in the block, the number after the slash is known as the mask, or size
- Let's break it down:

192.168.0.0/24

Starting IP Address Mask (Size)

CIDR RANGE TABLE

- The mask of a CIDR determines the size of the network or subnet thanks to binary and hexadecimal shenanigans
- Here's the short, cheat-sheet version:

CIDR Block	Mask	Number of Hosts
/14	255.252.0.0	262144
/15	255.254.0.0	131072
/16	255.255.0.0	65536
/17	255.255.128.0	32768
/18	255.255.192.0	16384
/19	255.255.224.0	8192
/20	255.255.240.0	4096
/21	255.255.248.0	2048
/22	255.255.252.0	1024
/23	255.255.254.0	512
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	32
/28	255.255.255.240	16
/29	255.255.255.248	8
/30	255.255.255.252	4

IN THE REAL WORLD

- As the chart shows, you can create a large VNet and/or subnets
- However, most of the time, you won't be creating networks or subnets that large
- Traditional network admins typically will create address spaces of a size you need, no more, no less
- This is especially true if there are going to be multiple VNets (there usually are) or multiple subnets (and there usually are)
- It's a good rule of thumb to know what your VNet and subnet requirements are:
 - For instance, a SQL Server Managed Instance requires a minimum of 14 IP addresses and can grow¹
 - Azure Databricks, on the other hand requires *two* subnets of the same size²
- Provide space for your services to grow, but don't get greedy

VIRTUAL NETWORK PEERING

- You might be wondering: “Can I just create one huge network, and just a bunch of subnets?”
- You COULD, but that’s not great; there may be reasons to separate networks out for security or manageability reasons
- Instead, it’s common to have multiple networks joined, or peered, together
- This allows network administrators to control traffic between the two using firewalls or other network devices
- We’ll see some common topologies that take advantage of peering later
- Some gotchas:
 - Your virtual network address spaces CANNOT overlap
 - You can peer both within a region or use global virtual network peering but be careful of traffic egress/ingress across regions!

WHAT DOES VIRTUAL NETWORKING COST?

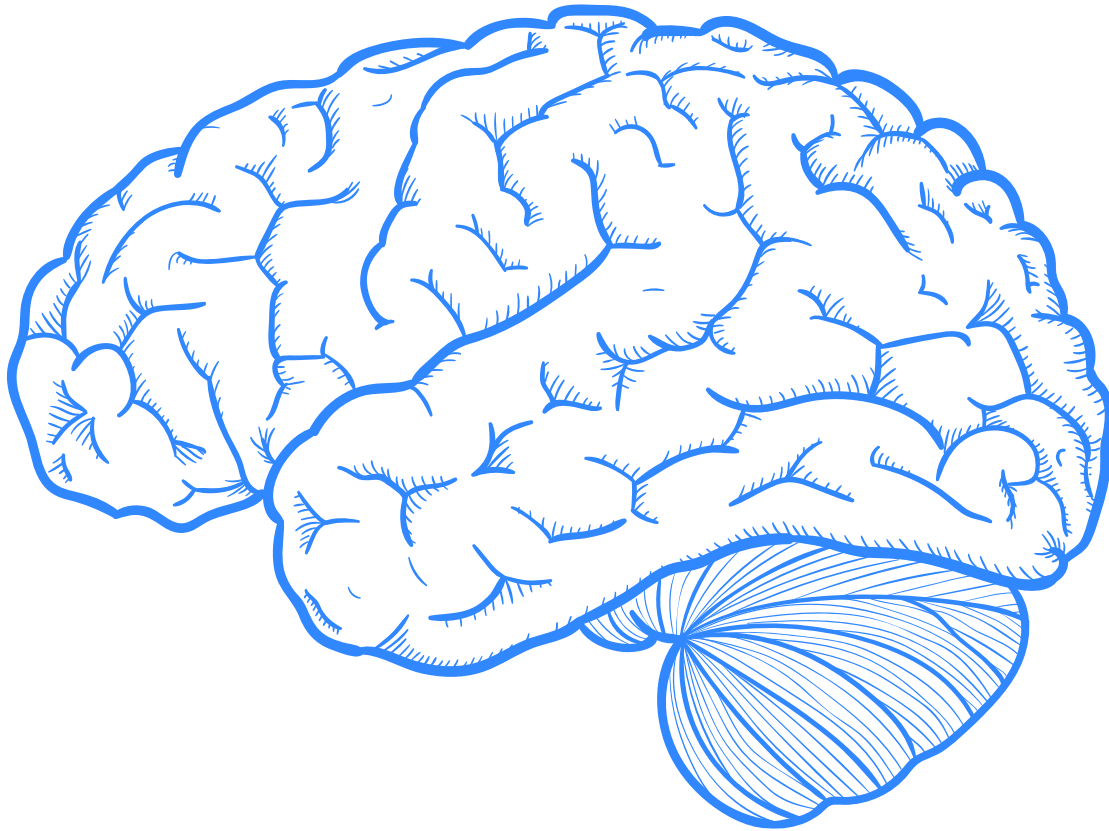
- Azure Virtual Networks don't cost anything...
- However, the services that send traffic across them can!
- There's also a cost for peered networks, even within the same region, and you're charged per gigabyte (GB) of traffic that enters and leaves a peered network.
- Costs vary between regions and peers
- Check out the pricing calculator for more information: <https://azure.microsoft.com/en-us/pricing/details/virtual-network/>

DEPLOYING A VIRTUAL NETWORK

- Once you have your address space and required subnet sizes, you can deploy a virtual network
- Can deploy via the portal, or through templates and automation (ARM, Bicep, Terraform, etc)
- Let's walk through a standard deployment now and explore the resource in Azure!



DEMO: DEPLOYING A VIRTUAL NETWORK



Pop quiz!

Based on what we talked about, consider the following:

- What purpose do Virtual Networks in Azure serve? Name as many as you can, and some reasons why someone or an organization may choose to deploy them.
- If you create a Virtual Network with a CIDR address space of 192.168.0.0/24, how many subnets can you create in that virtual network that have room for 64 addresses?
- True or false: If you have a virtual network with an address space of 192.168.0.0/26, and another virtual network of 192.168.0.64/26, can these virtual networks be peered?



AZURE SERVICES AND CONNECTIVITY

So just how DO we connect to services in Azure when we want to use them?

PUBLIC DOMAIN

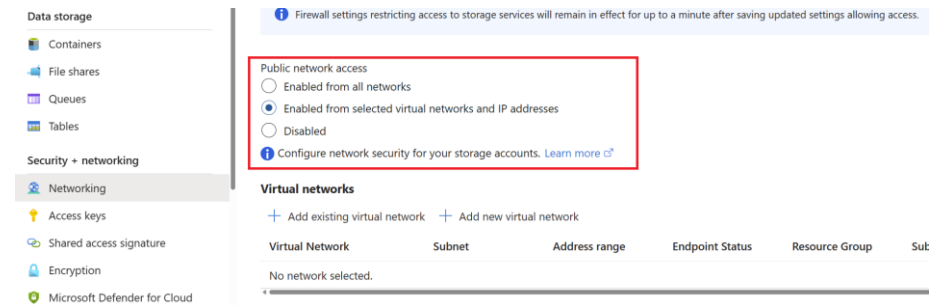
- Azure is a public cloud, and as such, the services you deploy in them are designed to be accessed from pretty much anywhere
- By default, most services you deploy into Azure have a public interface, aka, a public IP address which is resolved via some sort of fully qualified domain name (FQDN)
- This means that if you connect to the service, you're doing so over the public internet
- However, it's important to note that:
 - You still need to authenticate to the service
 - Your traffic is still encrypted
 - There may be per-service firewall or IP access lists that restrict access

IN THE REAL WORLD (AGAIN)

- A lot of organizations don't want the traffic going over the public internet, and would instead prefer to control the source of the traffic
 - Internal security policies
 - Regulatory requirements
- This is where we come back to our Azure VNets
 - We can control access to services from specific virtual networks
 - Service Endpoints
 - Service Tags
 - We can even provide private IPs to our services, further providing control of the traffic (Private Link)
- We can also effectively disable public access, provided we still have a way to connect afterwards

LIMITING ACCESS VIA VIRTUAL NETWORKS

- Certain services allow you to restrict access from certain virtual networks or IP ranges
- Enabling this feature means that you're allowing traffic from the specified locations
- For example, let's look at an Azure Storage Account:



- We can allow only traffic from virtual networks or subnets we want from here, or whitelist IP addresses
- This works, but it's a lot of overhead and upkeep

AZURE PRIVATE LINK

- We can also set up private IP addresses for our Azure services and endpoints
- This is called Azure Private Link
- Uses resources called private endpoints
 - Creates a network adapter which then receives a private IP address for the service
 - Users and other services can then use this private IP to connect instead of the public IP
- Must be deployed into a private network subnet
- Can have multiple endpoints for the same service
- Needs DNS resolution to work properly
 - Azure Private DNS
 - Custom DNS Servers

AZURE PRIVATE DNS ZONES

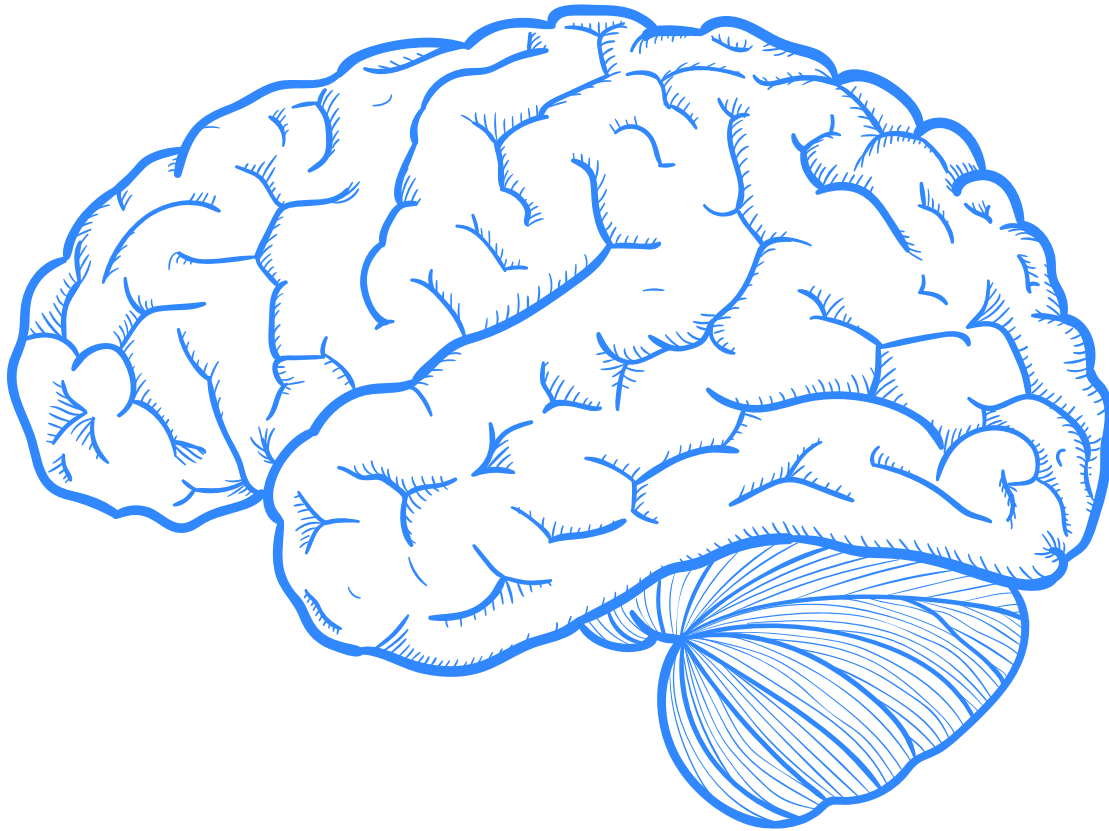
- This goes together with private endpoints
- If you deploy a private endpoint, you need some way to tie that IP address to a service
- Azure has Private DNS Zones for this
 - They are associated with a virtual network and provide resolution to resources that reference it, just like a DNS server
 - Stores FQDNs for a given domain (resource), e.g., privatelink.blob.core.windows.net
 - Need one zone per service type
 - Can have multiple, but harder to manage
 - Your services need visibility to this
- Alternatively, you can host your own DNS servers in Azure (or some other service), but the same rules still apply: somehow, a user or service HAS to be able to resolve the private IP via the hostname, or they'll get the public IP

NETWORK SECURITY GROUPS

- Are assigned to the subnet level of a virtual network
- Essentially a port-based firewall for allowing traffic in or out of a subnet
- Has default rules
- Specific rules may need added for services like Azure SQL, RDP, SSH, etc.
- Lack of an NSG = No rules

“HELP! I CAN’T CONNECT!”

- Once you start looking for ways to restrict access to Azure services, sooner or later, someone won’t be able to connect
- Most common error you’ll receive is a “403 – Forbidden” response
 - This is an HTTP response code
 - It means the request was denied by the service
 - Not permissions based!
- Not to be confused for “401 – Unauthorized” response, which is either due to insufficient (or incorrect) credentials’
- Make sure that the requestor:
 - Is resolving the correct address for the service (in the case of private link)
 - Has a “path” to the service via peering or endpoints
- If someone is getting a timeout, this usually means a network security group/firewall rule is in place preventing the traffic



Pop quiz!

Based on what we talked about, consider the following:

- A VM on a virtual network needs to connect to an Azure SQL Database that does not have a private endpoint defined. Name two possible ways that you can make sure that the virtual machine can connect to the SQL Database.
- A VM on a virtual network is trying to connect to an Azure SQL Database that has a private endpoint defined on a different virtual network, but you keep getting an error that the server has public access disabled. What are some possible reasons that the VM can't connect to the Server, if the private endpoint does exist for the server?
- What is the difference between private networking, and Azure backbone connections through whitelisted Virtual Networks?



DEMO: CONNECTIVITY AND AZURE SERVICES



FROM ON PREMESIS TO AZURE

Because cloud-to-cloud isn't enough in our hybrid world

GETTING FROM HERE TO THERE

- Extending your on-premises network(s) to Azure is usually done by your networking team and cloud administrators
- You should still be aware of how people are connecting, and what those options are:
 - Point to Site VPN
 - Site to Site VPN
 - Express Route
- Making sure you can connect to your services via private link is one of the hardest challenges you may face and potentially troubleshoot
- All of these configurations rely on what's called a VPN Gateway Resource
 - Needs a special subnet (Gateway Subnet) defined
 - Also sometimes requires a local network gateway (to let Azure know what your local address spaces are)

POINT TO SITE VPN

- Essentially “dial-up” for Azure
- You have a VPN client that connects to Azure, and once connected and authenticated, your local machine is “connected” to a virtual network
- Great for home labs or experimenting with Azure networking
- Supports many different encryption and authentication options
- Hard to scale
- Limited bandwidth

SITE TO SITE VPN

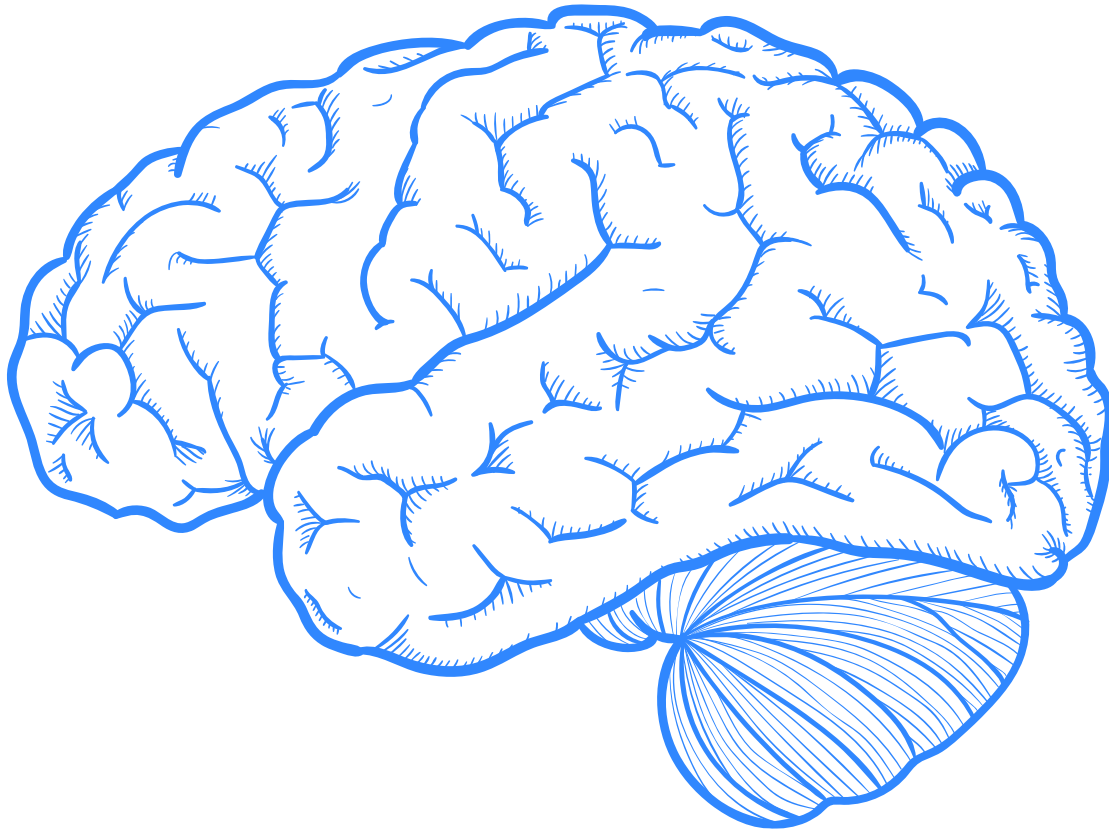
- Provides connectivity from on-premises via your data center's gateway/router
- “Always on” and provides a path to Azure via a secure VPN tunnel
- Great for home labs (if they have the right equipment) or small to medium businesses
- Better bandwidth than point to site, but still not the biggest pipe(s)
- Can be turned off and on as needed

EXPRESS ROUTE

- Direct circuit to Azure provided by carrier
- Highest (potential) bandwidth (up to 10Gbps!)
- Most expensive

HAVING CONNECTIVITY ISN'T THE ONLY PROBLEM

- Just having the route from on-premises isn't enough, your clients still need to resolve DNS hostnames correctly
- Just how DO you get DNS resolution working correctly?
- And how do you tell if it isn't? How can you test locally?
 - DNS Conditional Forwarding
 - Private DNS Resolver
 - Custom DNS Servers



Pop quiz!

Based on what we talked about, consider the following:

- You get an email/Teams/Slack message from a remote worker that yesterday they were able to connect to a storage account, but today they're getting a 403: Unauthorized in Azure Storage Explorer when attempting to upload some new data to a data lake. They said that yesterday they were in the office, and it worked fine. How would you troubleshoot this scenario?
- Your company is considering an Azure migration, and as part of that migration, you need to move several large (TB) databases to an Azure VM. What type of connectivity from on-premises should you recommend to make sure the data copies can be accomplished as fast as possible?



DEMO: SETTING UP ON PREMISES CONNECTIONS



COMMON AZURE NETWORK TOPOLOGIES

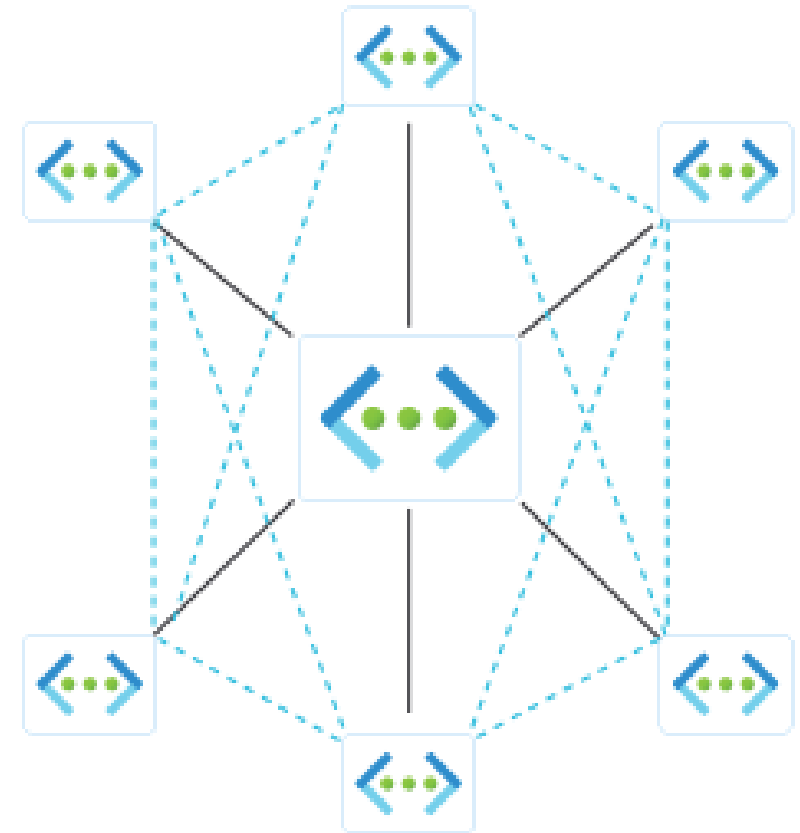
Be on the lookout for these common deployment models and how you play in them

REFERENCE PATTERNS TO THE RESCUE

- Without some kind of order, cloud deployments of all types can sprawl and become unmanageable quickly
- When it comes to the network level, there's couple common patterns to be aware of, and depending on which one is use in your environment will give you a lot of information how other users and services will connect to your services
- There can be different strategies to all accomplish the same goal
- Comes down to how the team(s) wants to manage the topology, and more importantly, secure and isolate the traffic

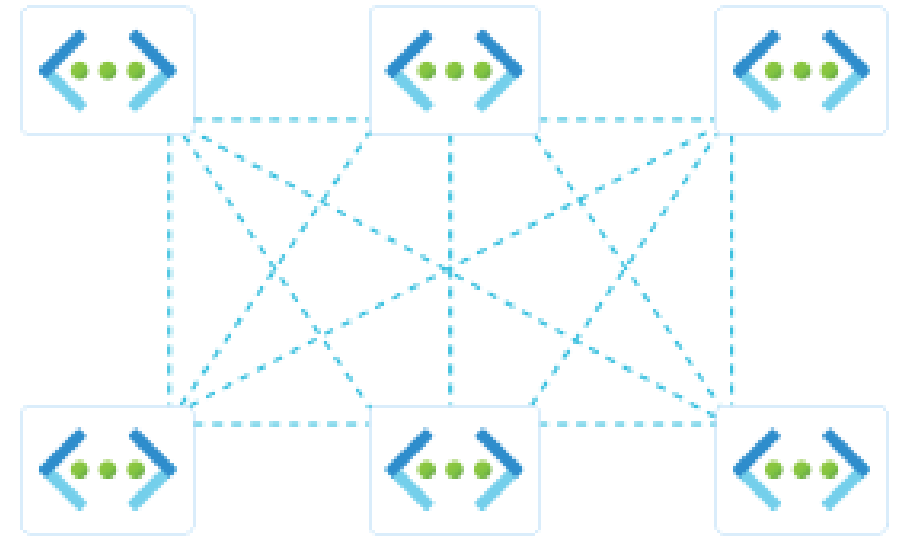
HUB AND SPOKE

- Hub and Spoke topologies are exactly what they sound like: one, central “hub” virtual network that is peered to different spokes
- Consider the diagram on the right: A central hub network where common network resources live, such as the firewall and gateways (for on premises connections), which is then peered to both a “data” network and “application” network for virtual machines or app services
- The spokes are designed to isolate services and networks from the other spokes
- All traffic must traverse the hub network, which makes it a good place for a firewall but can also introduce latency between services
- Spokes CAN be peered, but sometimes defeats the purpose...



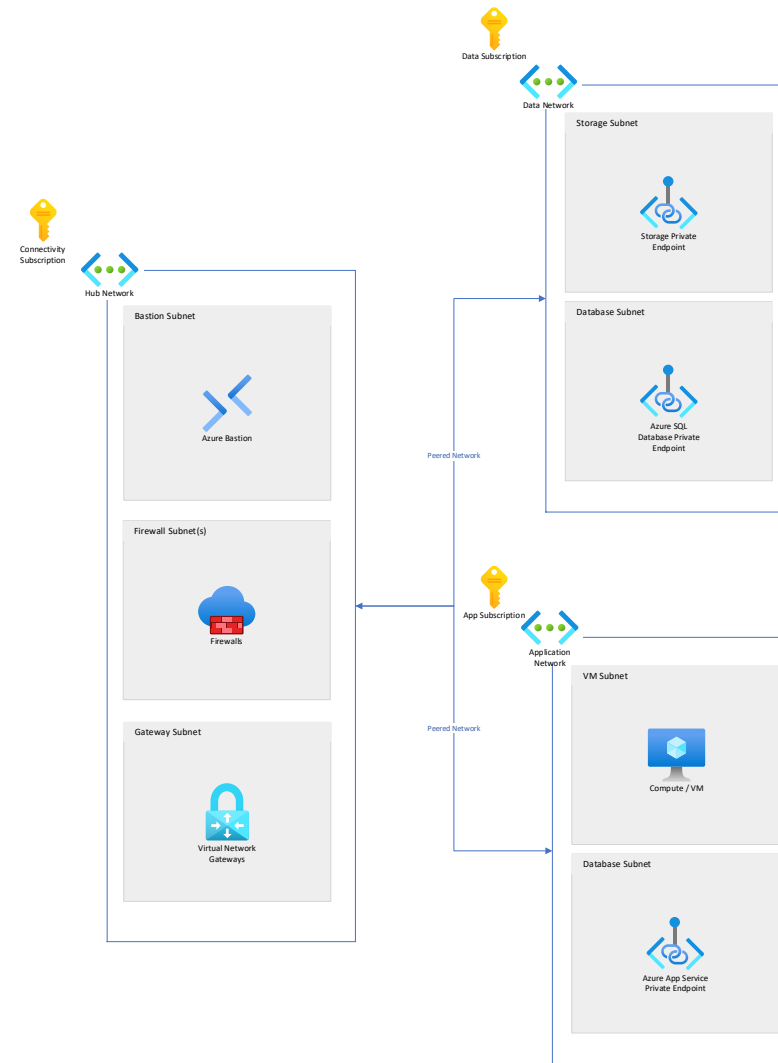
MESH NETWORKING

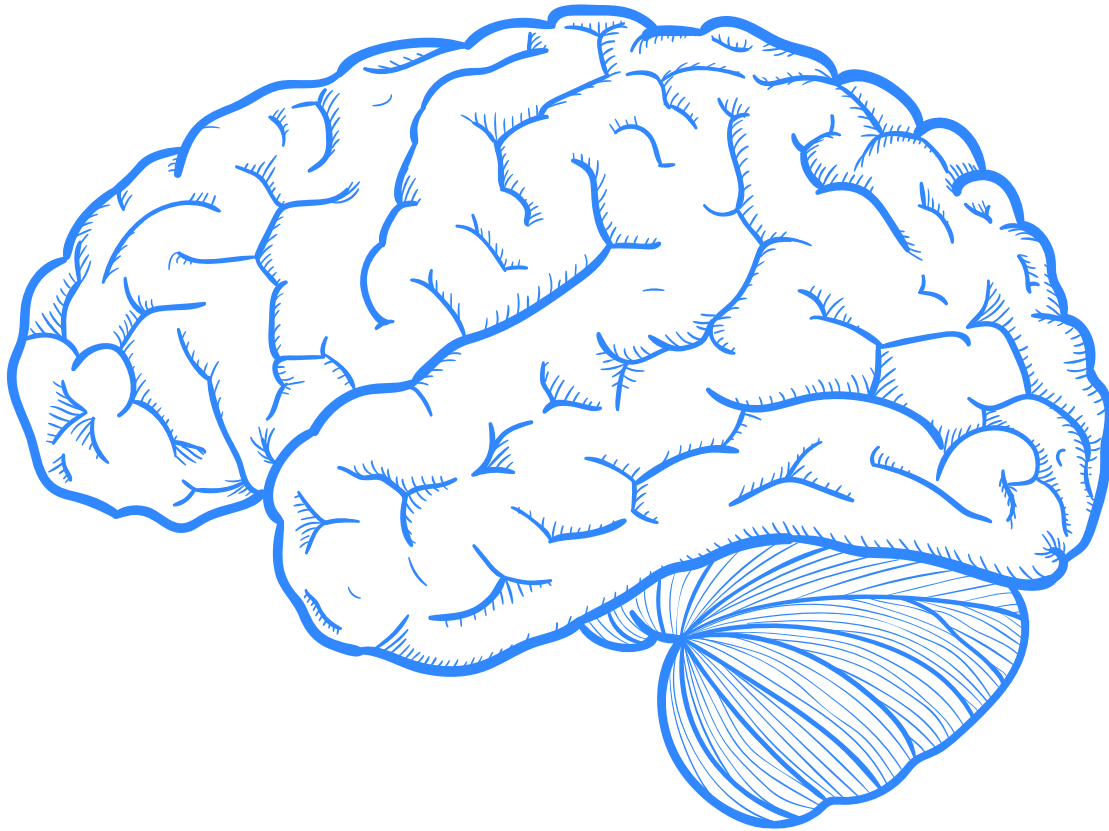
- All networks are peered to each other
- Reduces any throughput or latency from going through a hub network by providing direct connectivity to each network
- Slightly harder to manage shared resources (like Azure DNS Zones) to share between networks
- Peering limits



A HUB AND SPOKE EXAMPLE

- Consider a simple hub and spoke for a web application with data backend
- Isolated at the subscription level
- All traffic routes through hub network
- Think about this diagram a bit: what other resources might you want to deploy in each subscription and make part of each network?
- How would you segment each network?





Pop quiz!

Based on what we talked about, consider the following:

- Your company has a hub and spoke network architecture created, and you have a dedicated virtual network peered from the hub network where you host your SQL Servers and Azure SQL Databases, as well as several storage accounts. The web team has just started deploying their services in another spoke network, and they're having trouble reaching your services. What might be causing this issue, and what kind of actions can you recommend to help them connect?

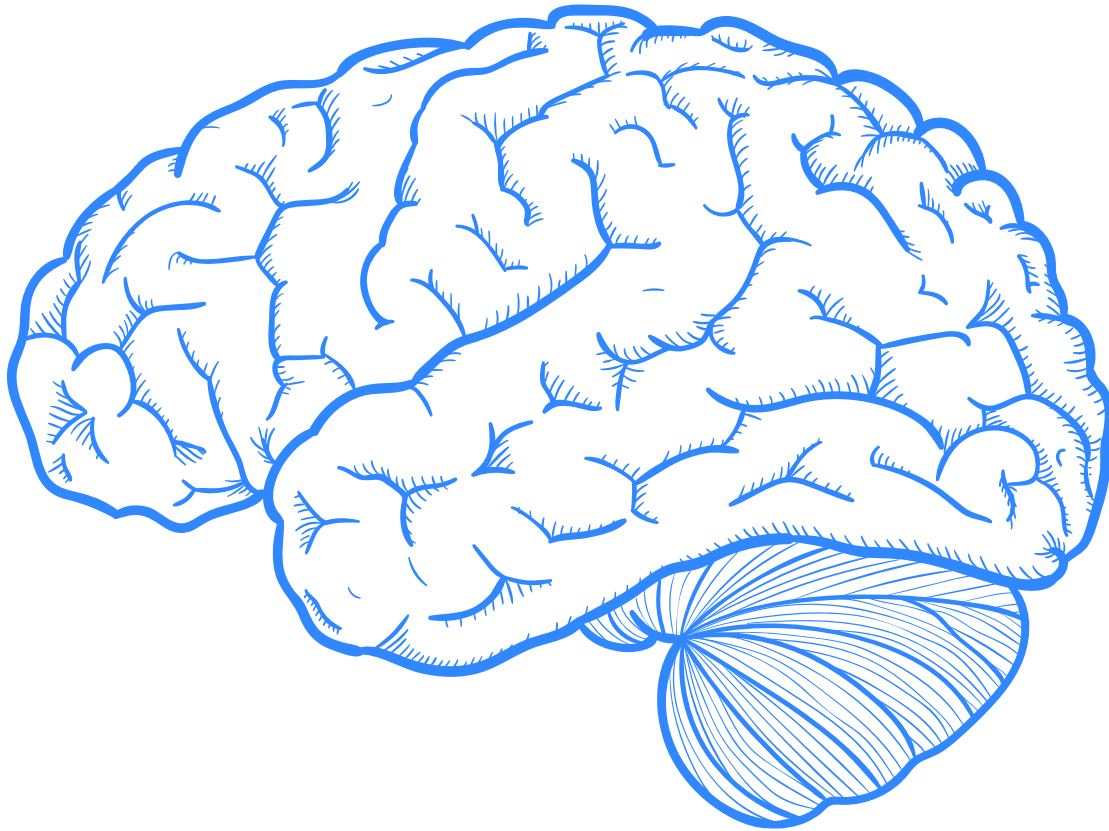


MANAGED VIRTUAL NETWORKS

Some services play by a different set of rules

WAIT, *ANOTHER* TYPE OF NETWORK?!

- If you plan on using Azure Private Link and Private Endpoints, you might soon discover some other Azures services can't connect to your services
- This is because many Azure services use public-based resources, and if your resources aren't publically accessible, then they won't work
- This is where Managed Virtual Networks come in
 - Certain Azure services, like Azure Data Factory, give you the option to use Managed Virtual Networks as part of the service
 - This creates an isolated, "unreachable" private network from which you can define additional private endpoints for your services that it needs
 - Bottom line: If you see this option, you'll probably need it if you're using some kind of private connectivity



Pop quiz!

Based on what we talked about, consider the following:

- You are creating some new data pipelines in Azure Data Factory, and you need to establish some private connections to your data services in Azure. You are using a Managed Virtual Network-enabled runtime, and you set up your managed private endpoints to the services. However, you still can't connect. What should you verify on your data services?



DEMO: MANAGED VIRTUAL NETWORKS EXAMPLE



WHAT'S NEXT?

Armed with this knowledge, where can you go to learn more?

NOW WHAT?

- If you're already in Azure today, how do you think about connectivity? Are you using private endpoints? Should you be?
- Or, if you're going to Azure soon, how does this impact your deployment plans?
- How will you test the connectivity?
- What are the networking requirements of the services you want to deploy?

RESOURCES TO READ

- Azure Virtual Networking Best practices: <https://learn.microsoft.com/en-us/azure/virtual-network/concepts-and-best-practices>
- Azure Private Link: <https://learn.microsoft.com/en-us/azure/private-link/private-link-overview>
- Azure Data Factory Managed Virtual Networks: <https://learn.microsoft.com/en-us/azure/data-factory/managed-virtual-network-private-endpoint>
- Azure Point To Site Configuration Walkthrough: <https://learn.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-howto-point-to-site-rm-ps>
- Slide Footnotes:
 - 1. Determine required subnet size and range for Azure SQL Managed Instance: <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/vnet-subnet-determine-size?view=azuresql-mi>
 - 2. Deploy Azure Databricks in your Azure virtual network (VNet injection): <https://learn.microsoft.com/en-us/azure/databricks/administration-guide/cloud-configurations/azure/vnet-inject>

THANK YOU!